European vs. American Options

Pricing Methods, Algorithms, and Numerical Examples

Equity Derivatives Library

Outline

1. European vs. American Options
2. Pricing Methods
3. Black-Scholes (European)
4. Monte Carlo Simulation
5. American Option Pricing
6. Numerical Examples and Benchmarking
7. Summary

What is an Option?

An option gives the holder the **right, but not the obligation**, to buy (*call*) or sell (*put*) an underlying asset at a fixed **strike price** K.

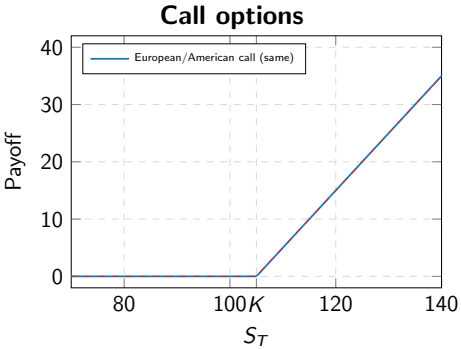
European Option

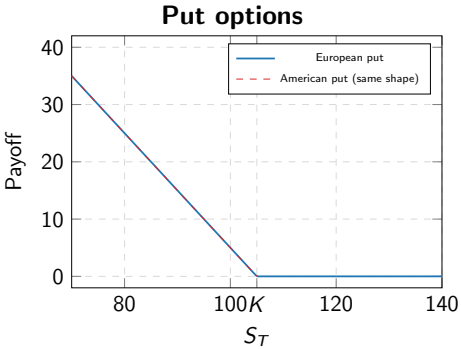
- May only be exercised **at maturity** T
- Simpler to price analytically (Black-Scholes)
- Call payoff: $\max(S_T - K, 0)$
- Put payoff: $\max(K - S_T, 0)$

American Option

- May be exercised **at any time** $t \leq T$
- Early exercise can be optimal for puts (and calls on dividend-paying stocks)
- Same payoff formulas, but evaluated at the chosen exercise time
- Always worth **at least as much** as the European counterpart

Payoff Diagrams at Expiry





The *terminal* payoff is identical—the difference lies in **when** exercise is allowed.

Early Exercise and the Early-Exercise Premium

When is early exercise optimal?

- **American call** on a *non-dividend-paying* stock: **never** optimal to exercise early $\Rightarrow$ equals the European call price.
- **American put**: early exercise can be optimal when the option is deep in-the-money. Receiving $K - S$ now and investing at rate r can outweigh waiting.

Early-exercise premium $= C^{\text{Am}} - C^{\text{Eu}} \geq 0$

Parameters: $S_0 = 100$, $K = 105$, $T = 1$, $r = 5\%$, $\sigma = 20\%$

	European	American	Premium
Call	8.0214	8.0232	≈ 0 (no dividends)
Put	7.9004	8.7416	**+0.8412**

Overview of Pricing Methods

Method	Option type	Complexity	Accuracy
Black-Scholes formula	European only	$O(1)$	Exact (GBM)
Binomial tree (CRR)	European & American	$O(N^2)$	Converges in N
Monte Carlo (plain)	European & exotic	$O(M)$	$\pm 1/\sqrt{M}$
Longstaff-Schwartz (LSM)	American	$O(M \cdot N)$	$\pm 1/\sqrt{M}$

- N = number of time steps, M = number of Monte Carlo paths
- This library implements **all four** methods
- Black-Scholes and binomial trees serve as **benchmarks** for validating Monte Carlo results

Black-Scholes Model

Under the **risk-neutral measure**, stock prices follow GBM:

$$dS = rS\,dt + \sigma S\,dW^{\mathbb{Q}}$$

Closed-form prices

$$C = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2), \qquad P = Ke^{-rT}\Phi(-d_2) - S_0\Phi(-d_1)$$

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}$$

Key properties

- Put-call parity: $C - P = S_0 - Ke^{-rT}$
- Only applies to **European** options on non-dividend-paying stocks
- Benchmark for Monte Carlo validation

Black-Scholes: Numerical Example

Parameters: $S_0 = 100,\ K = 105,\ T = 1\,\text{yr},\ r = 5\%,\ \sigma = 20\%$

Calculation

$$d_1 = \frac{\ln(100/105) + (0.05 + 0.02)}{0.20} = 0.0217$$

$$d_2 = 0.0217 - 0.20 = -0.1783$$

$$\Phi(d_1) = 0.5087, \quad \Phi(d_2) = 0.4292$$

$$C = 100(0.5087) - 105e^{-0.05}(0.4292)$$
$$= \mathbf{8.0214}$$

$$P = 105e^{-0.05}(0.5708) - 100(0.4913)$$
$$= \mathbf{7.9004}$$

Put-call parity check

$$C - P = 8.0214 - 7.9004 = 0.1210$$

$$S_0 - Ke^{-rT} = 100 - 105e^{-0.05} = 0.1209 \checkmark$$

Monte Carlo validation

	MC price	BS price
Call	8.0398	8.0214
Put	7.9214	7.9004

(200k paths, SE ≈ 0.03)

Monte Carlo Pricing (European Options)

Algorithm

1. Simulate M terminal stock prices under $\mathbb{Q}$:

$$S_T^{(i)} = S_0 \exp\Big[\big(r - \tfrac{\sigma^2}{2}\big)\,T + \sigma\sqrt{T}\,Z^{(i)}\Big], \quad Z^{(i)} \sim \mathcal{N}(0,1)$$

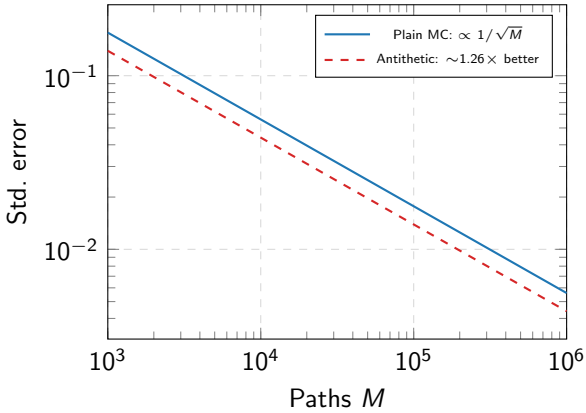
2. Compute payoffs: $h^{(i)} = \max(S_T^{(i)} - K,\, 0)$
3. Estimate price: $\hat{C} = e^{-rT}\,\dfrac{1}{M}\sum_{i=1}^{M} h^{(i)}$

Standard error scales as $1/\sqrt{M}$: doubling accuracy costs $4\times$ more paths.

Variance reduction: antithetic variates

For each $Z^{(i)}$, also simulate with $-Z^{(i)}$. Average the paired payoffs to cancel common noise. In practice: **$\sim$1.26$\times$ SE reduction** (free, no extra simulations).

Monte Carlo: Standard Error vs. Path Count



European call, $K = 105$, $T = 1$yr

Paths	Std. error	95% CI width
10,000	0.054	± 0.11
50,000	0.024	± 0.05
100,000	0.017	± 0.03
200,000	0.012	± 0.02
1,000,000	0.005	± 0.01

Antithetic variates (200k paths): SE $= 0.0236$ vs. 0.0297 plain.

CRR Binomial Tree (American Options)

Setup (Cox-Ross-Rubinstein, 1979)
Divide $[0, T]$ into N steps of size $\Delta t = T/N$:

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = 1/u, \quad p = \frac{e^{r\Delta t} - d}{u - d}$$

Algorithm

1. Build terminal node prices $S_0\, u^j\, d^{N-j}$
2. Set terminal values $V_j = \max(K - S_j, 0)$
3. Roll back:

$$V_j = e^{-r\Delta t}(p\, V_j + (1-p)\, V_{j+1})$$

4. Apply early-exercise:

$$V_j \leftarrow \max(V_j,\ K - S_j)$$

Convergence (American put, $K = 105$)

Steps N	Price
10	8.8550
50	8.7357
100	8.7476
200	8.7436
500	**8.7416**

Converges to ≈ 8.742 well before $N = 500$.

Longstaff-Schwartz (LSM) Algorithm

Key idea: use least-squares regression to estimate the *continuation value* at each exercise date.

Algorithm (Longstaff & Schwartz, RFS 2001)

1. Simulate M full price paths $\{S_t^{(i)}\}$, $t = 0, \Delta t, \ldots, T$
2. Initialise $V^{(i)} = \max(K - S_T^{(i)}, 0)$ (terminal payoff)
3. **Backward induction**: for $t = T - \Delta t, \ldots, \Delta t$:
   1. Discount: $V^{(i)} \leftarrow e^{-r\Delta t}\, V^{(i)}$
   2. For in-the-money paths, regress $V^{(i)}$ on $\left[1,\, S_t^{(i)},\, (S_t^{(i)})^2\right]$ to estimate the continuation value $\hat{C}(S_t^{(i)})$
   3. If $\max(K - S_t^{(i)}, 0) > \hat{C}(S_t^{(i)})$: exercise early, set $V^{(i)} = K - S_t^{(i)}$
4. Price $= e^{-r\Delta t}\, \mathbb{E}[V]$

LSM: Code Sketch

```
# Simulate M paths with N steps
paths = simulator.simulate_paths(T, N, M, risk_neutral=True)

# Terminal payoff
V = np.maximum(K - paths[:, -1], 0.0)

# Backward induction
for step in range(N-1, 0, -1):
    V *= discount                # discount one period
    S = paths[:, step]
    intrinsic = np.maximum(K - S, 0.0)
    itm = intrinsic > 0          # in-the-money mask

    # Least-squares regression on ITM paths
    X = np.column_stack([np.ones(itm.sum()), S[itm], S[itm]**2])
    coeffs, *_ = np.linalg.lstsq(X, V[itm], rcond=None)
    continuation = X @ coeffs

    # Early exercise where intrinsic beats continuation
    exercise = intrinsic[itm] > continuation
    V[np.where(itm)[0][exercise]] = intrinsic[itm][exercise]

V *= discount                    # discount to t=0
price = V.mean()
```

Numerical Results: All Methods

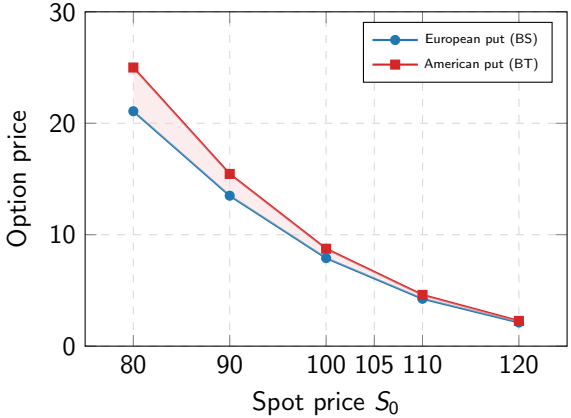
Parameters: $S_0 = 100$, $K = 105$, $T = 1\,\text{yr}$, $r = 5\%$, $\sigma = 20\%$

Option	Method	Price	Std. error	vs. benchmark
European call	Black-Scholes	8.0214	exact	—
	Monte Carlo	8.0398	0.0297	+0.018
	Antithetic MC	8.0461	0.0236	+0.025
European put	Black-Scholes	7.9004	exact	—
	Monte Carlo	7.9214	0.0232	+0.021
American call	Binomial ($N = 500$)	8.0232	—	—
	LSM ($M = 200$k)	8.0113	0.0294	−0.012
American put	Binomial ($N = 500$)	8.7416	—	—
	LSM ($M = 200$k)	8.6972	0.0182	−0.044

All MC differences lie within $\sim$2 standard errors of the benchmark.

Early-Exercise Premium vs. Moneyness

American put premium $= C_{\text{put}}^{\text{Am}} - C_{\text{put}}^{\text{Eu}}$ as a function of spot price ($K = 105$, $T = 1$yr, $r = 5\%$, $\sigma = 20\%$)



S_0	EU put	AM put	Premium
80	21.079	**25.000**	**3.921**
90	13.500	15.454	1.954
100	7.900	8.742	0.841
110	4.249	4.600	0.351
120	2.122	2.263	0.141

Premium is largest when the option is **deep in-the-money**: immediate exercise yields $K - S_0$ cash now, which earns interest. At $S_0 = 80$, the American put equals its intrinsic value $K - S_0 = 25$ (immediate exercise is optimal).

Benchmarking: LSM vs. Binomial Tree

American put ($S_0 = 100,\ K = 105,\ T = 1\text{yr}$)

Method	Price	Error
BT $N = 500$ (ref.)	8.7416	—
BT $N = 10$	8.8550	$+0.113$
BT $N = 50$	8.7357	-0.006
BT $N = 100$	8.7476	$+0.006$
BT $N = 200$	8.7436	$+0.002$
LSM $M = 50$k	8.71	-0.03
LSM $M = 100$k	8.72	-0.02
LSM $M = 200$k	8.6972	-0.044

Observations

- Binomial tree converges quickly; $N = 50$ is already accurate to < 0.01
- LSM is a **lower bound**: regression underestimates continuation value, triggering slightly too much early exercise
- Both methods agree to within ± 0.05 — well within Monte Carlo standard error (~ 0.02–0.03)
- Binomial tree is fast for single options; LSM scales to **path-dependent** and **multi-asset** settings where trees are infeasible

Summary

Key takeaways

- European options: exercisable only at T; priced exactly by **Black-Scholes**
- American options: exercisable at any $t \leq T$; always $\geq$ European counterpart
- American call $\approx$ European call (no dividends)
- American put carries a meaningful **early-exercise premium**, largest when deep ITM
- **LSM** extends Monte Carlo to American options via regression-based continuation value estimation
- **Binomial trees** give fast, accurate benchmarks for vanilla American options

Method comparison

	European	American
Analytical	BS	—
Tree	CRR	CRR + EE
Monte Carlo	Plain MC	LSM

This library

- `engine.price()` — European / exotic
- `engine.price_american()` — LSM
- `black_scholes_call/put` — BS benchmark
- `binomial_tree_american_*` — tree benchmark

Thank you